

Activity 10.2: Boundary Conditions and Geological Meaning

Geophysics — Numerical Heat Flow

Purpose

To understand how boundary conditions encode geological assumptions in numerical models.

Boundary Conditions

Common thermal boundary conditions include:

- Fixed temperature (Dirichlet).
- Fixed heat flux (Neumann).
- Mixed or radiative conditions.

Tasks

1. Provide a geological example appropriate for each type of boundary condition.
2. Explain how prescribing surface temperature versus surface heat flow changes model behavior.
3. Compare boundary conditions appropriate for:
 - continental lithosphere,
 - oceanic lithosphere,
 - cooling magma bodies.
4. Discuss why inappropriate boundary conditions can produce physically misleading results even if the numerical solution is stable.

Reflection

Why are boundary conditions often the most geologically important part of a numerical model?